

The scalar-range completely bounded question is affirmative

Literature-already-answered note for arXiv:2011.10422

13 August 2026

Status. Answered affirmatively by Hartz–McCarthy, arXiv:2606.02922. If the antilinear map in the completely bounded conjecture takes values in the scalar multiples of the identity, then the desired sharp bound $\|\theta\|_{cb} \leq 2$ follows. This answers the explicit “we do not know” question on source PDF page 9. It does *not* settle either of the source paper’s two unrestricted abstract conjectures.

Source question

Clouâtre–Ostermann–Ransford formulate the completely bounded conjecture for a unital completely bounded homomorphism θ and a unital antilinear complete contraction α . Immediately after their $1 + \sqrt{2}$ result, they ask whether the sharp constant 2 follows for the disk algebra when α takes values in $\mathbb{C}1$ [1, p. 9].

Conjecture 6.1. *Let A be an operator algebra, let $\theta : A \rightarrow M_n(\mathbb{C})$ be a unital completely bounded homomorphism, and let $\alpha : A \rightarrow A$ be a unital antilinear complete contraction. Define a linear map $\theta_\alpha : A \rightarrow M_n(\mathbb{C})$ by (1). If $\|\theta_\alpha\|_{cb} \leq 1$, then $\|\theta\|_{cb} \leq 2$.*

We start by showing that the estimate $\|\theta\|_{cb} \leq 1 + \sqrt{2}$ always holds, mirroring the previous setting.

Proposition 6.2. *Let A be an operator algebra. Let θ, α and θ_α be as in Conjecture 6.1. Then $\|\theta\|_{cb} \leq 1 + \sqrt{2}$.*

Proof. Fix a positive integer n . Remark (i) following the proof of Theorem 1.1 shows that the argument therein applies to the maps $\theta^{(n)}$ and $\alpha^{(n)}$. Hence $\|\theta^{(n)}\| \leq 2$ for every $n \geq 1$, which is equivalent to $\|\theta\|_{cb} \leq 2$. $\square$

We note that $*$ -homomorphisms on C^* -algebras are always completely contractive, so that Theorem 1.5 shows that Conjecture 6.1 is verified for commutative C^* -algebras. In contrast, we do not know if the proof of Theorem 1.4 can be adapted to show that Conjecture 6.1 holds in the case of the disk algebra when α takes values in the scalar multiples of the identity.

Source evidence: arXiv:2011.10422v3, PDF page 9.

Proof intuition

When α has scalar range, its contribution to the symmetrized map is not an operator-valued perturbation at all: it is a scalar functional times the identity. Hartz and McCarthy prove that adding any such scalar perturbation to a unital homomorphism cannot reduce any matrix-amplified norm below the

homomorphism norm (unless that norm was already at most one). Applying this comparison at every matrix level turns the hypothesis on θ_α directly into the bound 2.

The comparison theorem is powered by an extremal-vector argument. Near a norm-attaining sequence for a homomorphism, scalar matrix perturbations are first-order orthogonal to the image. Potapov–Möbius automorphisms of the operator ball supply the admissible variations; taking a first variation removes the scalar cross term.

Theorem 1 (Hartz–McCarthy [2]). *Let $\mathcal{A}$ be a unital operator algebra, let $\theta : \mathcal{A} \rightarrow B(\mathcal{H})$ be a continuous unital homomorphism, and let $\beta \in \mathcal{A}^*$. For every $m \geq 1$,*

$$\|\theta^{(m)}\| \leq \max\left\{1, \|\theta^{(m)} + \beta^{(m)}I\|\right\}. \quad (1)$$

Consequently,

$$\|\theta\|_{cb} \leq \max\{1, \|\theta + \beta I\|_{cb}\}.$$

Why the comparison holds. Fix m , put $C = \|\theta^{(m)}\| > 1$, and choose contractions $a_k \in M_m(\mathcal{A})$ and unit vectors x_k with $\|R_k x_k\| \rightarrow C$, where $R_k = \theta^{(m)}(a_k)$. For a scalar matrix $D \in M_m(\mathbb{C})$, the Potapov–Möbius transform

$$M_D(X) = (I - DD^*)^{-1/2}(X + D)(I + D^*X)^{-1}(I - D^*D)^{1/2}$$

preserves the operator unit ball. The homomorphism identity

$$M_{D \otimes I}(R_k) = \theta^{(m)}(M_{D \otimes I}(a_k))$$

therefore bounds these variations by C . Expanding at tD to first order and using $\|R_k x_k\| \rightarrow C$ gives, uniformly for bounded D_k ,

$$\langle R_k x_k, (D_k \otimes I)x_k \rangle \rightarrow 0. \quad (2)$$

Taking $D_k = \beta^{(m)}(a_k)$, expanding the squared norm of $(\theta^{(m)} + \beta^{(m)}I)(a_k)x_k$, and using (2) yields

$$\|\theta^{(m)} + \beta^{(m)}I\|^2 \geq C^2.$$

This proves (1); the case $C \leq 1$ is immediate.

Application to the source question

Assume $\alpha(\mathcal{A}) \subset \mathbb{C}1$. Write $\alpha(f) = \lambda(f)1$, where λ is antilinear, and define the linear functional $\beta(f) = \overline{\lambda(f)}$. Because θ is unital,

$$2\theta_\alpha(f) = \theta(f) + \theta(\alpha(f))^* = \theta(f) + \beta(f)I.$$

Thus $\|\theta_\alpha\|_{cb} \leq 1$ implies $\|\theta + \beta I\|_{cb} \leq 2$. The theorem gives

$$\boxed{\|\theta\|_{cb} \leq 2}.$$

This argument works for every unital operator algebra, not only the disk algebra, and the comparison theorem itself does not require $\alpha(1) = 1$.

Exact remaining scope

The unrestricted norm and completely bounded conjectures in [1] allow operator-valued $\alpha(f)$, so the reduction to a scalar functional fails. Lorist–Schwenninger [3] prove the classical Crouzeix conjecture and an abstract variant under the stronger condition that $\frac{1}{2}(\theta + \theta \circ \alpha^*)$ is completely positive. They explicitly note that their argument does not directly yield the completely bounded Crouzeix statement. Hence those 2026 results answer the scalar-range question recorded here and the classical consequence, but they do not claim the source paper’s unrestricted abstract conjectures.

Provenance. This packet records a direct literature answer, not new mathematics from the run. Exact arXiv-id, title, author, formula, and core-keyword searches through 13 August 2026 found no earlier run packet for this source-to-answer match.

References

- [1] R. Clouâtre, M. Ostermann, and T. Ransford, *An abstract approach to the Crouzeix conjecture*, arXiv:2011.10422v3 (2020). See Conjecture 6.1 and the question on PDF page 9.
- [2] M. Hartz and J. E. McCarthy, *From Clouatre–Ostermann–Ransford to Okubo–Ando*, arXiv:2606.02922 (2026). See Theorem 1.2 and Section 4.
- [3] E. Lorist and F. Schwenninger, *A solution to Crouzeix’s conjecture*, arXiv:2608.03841 (2026). See Theorem 2.3 and Remark 2.5.